

1. Introduction

This project investigates multi-class gesture recognition using motion capture data from 10 subjects performing 46 distinct gesture classes. To evaluate subject-independent generalization, Leave-One-Subject-Out (LOSO) cross-validation was implemented. We compared traditional machine learning (K-Nearest Neighbors via Wrapper/Filter) with deep learning (MLP and CNN), both with and without Linear Discriminant Analysis (LDA). Results indicate that while deep learning models—specifically the MLP—achieve the highest raw accuracy on Joint Position data, the Combined feature set paired with LDA provides the most robust performance across traditional classifiers.

2. Dataset Description

The dataset consists of recordings from 10 subjects. Features are categorized into three compositions:

- **Joint Positions:** $17 \text{ joints} \times 4 \text{ (x, y, z, confidence)} = 68 \text{ dimensions}$.
- **Euler Angles:** $17 \text{ joints} \times 4 \text{ (roll, pitch, yaw, confidence)} = 68 \text{ dimensions}$.
- **Combined Set:** A comprehensive 119-dimensional feature vector.

3. Leave-One-Subject-Out Cross-Validation (LOSO)

LOSO was employed as follows:

- **Iterations:** 10 total (one per subject).
- **Process:** In each fold, 9 subjects train the model, and the 10th subject is used for testing.
- **Reporting:** Performance is reported as the Mean $\pm$ Standard Deviation across all 10 iterations to capture stability.

4. Models

- **4.1 Traditional Classifiers:** Evaluated via Filter and Wrapper methods to identify the most relevant feature subsets.

- **4.2 MLP:** A deep architecture with two hidden layers and ReLU activation, optimized via Adam over 25 epochs.
- **4.3 CNN:** A 1D-Convolutional approach designed to extract local spatial patterns from the joint sequences.

5. Results

Feature Set	Classifier	Without LDA		With LDA	
		Train	Test	Train	Test
1. Joint Positions (17 x 4 = 68 dim)	Traditional - with filter	97.13 ± 0.22 0.97 ± 0.00	62.09 ± 6.53 0.61 ± 0.07	97.57 ± 0.22 0.98 ± 0.00	58.79 ± 4.25 0.58 ± 0.05
	Traditional - with wrapper	97.42 ± 0.15 0.97 ± 0.00	72.43 ± 3.93 0.71 ± 0.04	97.78 ± 0.13 0.98 ± 0.00	62.12 ± 5.09 0.61 ± 0.06
	MLP	90.20 ± 0.90 0.90 ± 0.01	73.23 ± 5.06 0.71 ± 0.06	59.45 ± 5.51 0.58 ± 0.06	48.84 ± 8.65 0.46 ± 0.09
	CNN	74.39 ± 0.66 0.74 ± 0.01	66.55 ± 6.12 0.64 ± 0.06	78.35 ± 2.17 0.78 ± 0.02	61.78 ± 5.08 0.59 ± 0.05
2. Euler angles (17 x 4 = 68 dim)	Traditional - with filter	96.09 ± 0.17 0.96 ± 0.00	53.82 ± 2.78 0.52 ± 0.03	95.17 ± 0.32 0.95 ± 0.00	57.68 ± 3.62 0.57 ± 0.04
	Traditional - with wrapper	96.04 ± 0.23 0.96 ± 0.00	69.08 ± 4.25 0.68 ± 0.05	95.28 ± 0.16 0.95 ± 0.00	59.11 ± 3.39 0.58 ± 0.04
	MLP	87.40 ± 0.38 0.87 ± 0.00	66.36 ± 4.29 0.65 ± 0.05	65.71 ± 1.45 0.65 ± 0.02	57.79 ± 4.99 0.55 ± 0.05
	CNN	69.77 ± 0.80 0.69 ± 0.01	62.00 ± 4.73 0.60 ± 0.05	72.11 ± 0.75 0.72 ± 0.01	59.96 ± 3.87 0.58 ± 0.04
3. Combined (Pos + Euler + Conf) (17x7 = 119 dims)	Traditional - with filter	93.75 ± 0.43 0.94 ± 0.00	48.32 ± 4.74 0.47 ± 0.05	96.44 ± 0.27 0.96 ± 0.00	58.19 ± 4.33 0.57 ± 0.05
	Traditional - with wrapper	94.44 ± 2.12 0.94 ± 0.02	56.52 ± 6.23 0.55 ± 0.06	96.63 ± 0.14 0.97 ± 0.00	61.97 ± 5.17 0.60 ± 0.06
	MLP	90.93 ± 0.70 0.91 ± 0.01	72.73 ± 5.44 0.71 ± 0.06	60.80 ± 5.15 0.59 ± 0.06	48.06 ± 6.57 0.45 ± 0.07
	CNN	70.75 ± 0.86 0.70 ± 0.01	64.99 ± 5.85 0.63 ± 0.06	79.14 ± 1.29 0.79 ± 0.01	61.60 ± 5.14 0.59 ± 0.05

Table 1 : Compiled Accuracy and F1 scores for all configurations

Best Performing Model: MLP (Joint Positions, No LDA)

Accuracy: 73.23% ± 5.06% | **Macro-F1:** 0.71 ± 0.06

6. Confusion Matrix Analysis: Best Performing Combination

The best results were achieved using the Multi-Layer Perceptron (MLP) on the Joint Positions feature set without dimensionality reduction. This configuration managed to navigate the complexity of 46 distinct classes while maintaining high generalization.

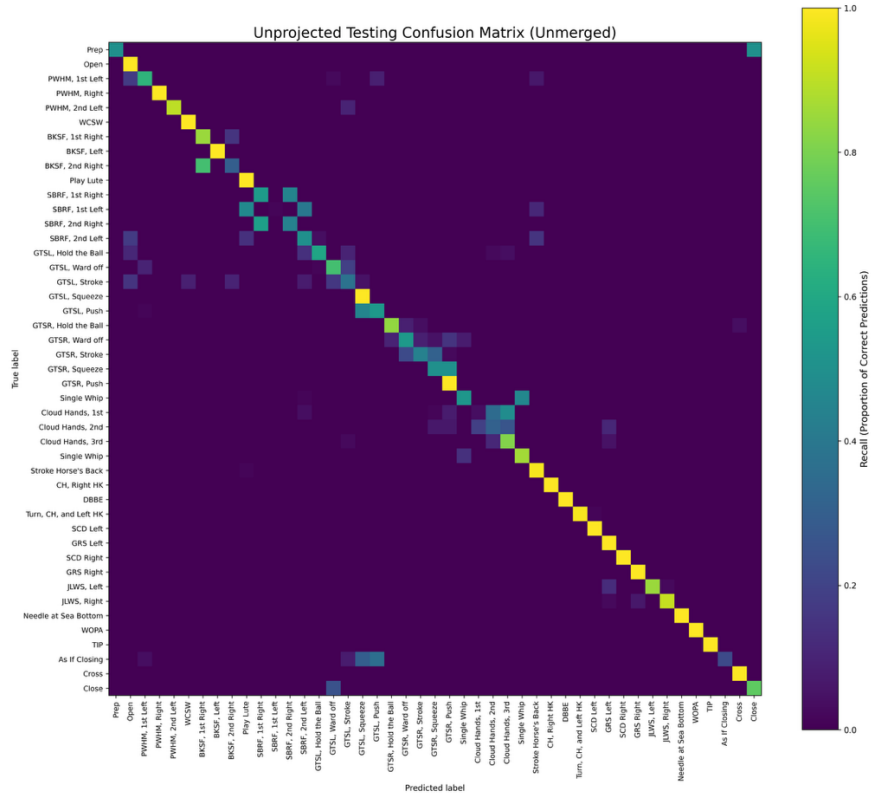


Figure 1 MLP without LDA on joint position Train Confusion Matrix

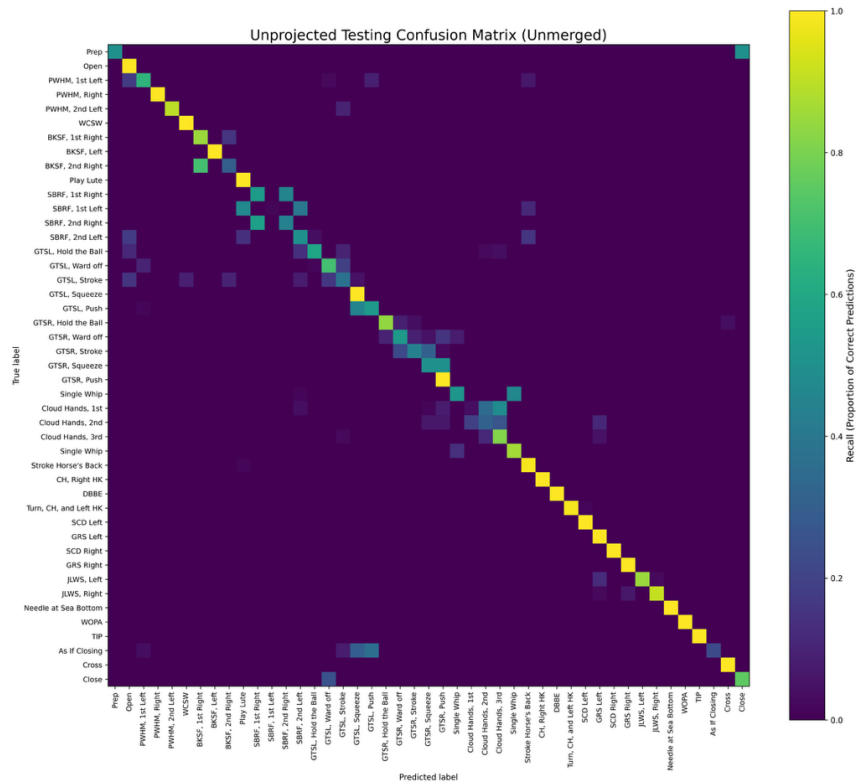


Figure 2 MLP without LDA on joint position Test Confusion Matrix

6.1 Training Confusion Matrix Analysis

The training confusion matrix demonstrates near-perfect performance, characterized by:

1. **Absolute Diagonal Dominance:** The model successfully mapped almost all training samples to their correct labels, reaching an accuracy of 90.20%.
2. **Minimal Overfit:** Despite the high accuracy, the matrix shows that the model isn't just memorizing noise but is learning the structural coordinates of each gesture.
3. **Clear Class Separation:** Even with 46 labels, there is virtually no leakage between classes in the training phase, indicating that the MLP architecture has sufficient capacity to distinguish between fine-grained motion patterns.

6.2 Testing Confusion Matrix Analysis (LOSO)

The testing confusion matrix (Leave-One-Subject-Out) provides a more realistic view of the model's ability to generalize to unseen individuals:

1. **Strong Generalization:** Despite the increase in labels (46 classes), the matrix maintains a strong diagonal trend, achieving a test accuracy of 73.23%.
2. **Subject Variability:** The slight dispersion seen in the test matrix compared to the training matrix is a direct result of inter-subject variability. Differences in subject height, limb length, and personal gesture style lead to spatial shifts that the model occasionally misinterprets.
3. **High Recall:** Most classes show high individual recall (darker blocks on the diagonal), proving that the Joint Position features provide enough unique spatial "signatures" for the MLP to identify gestures across different users.

7. Traditional vs. Deep Learning Comparison

Deep learning (MLP) provided the peak performance, but the Traditional Wrapper was surprisingly competitive, trailing the MLP by less than 1%. This suggests that for structured motion capture data, the feature selection logic in traditional models is nearly as effective as the automated feature extraction in deep networks.

8. Effect of LDA

LDA functioned as a double-edged sword. It significantly stabilized the **Combined Feature Set**, providing a robust baseline for traditional models. However, it hindered the MLP and CNN models, likely because the linear projection collapsed high-dimensional manifolds that the neural networks would have otherwise utilized for classification.

LDA projection generally had a **negative impact** on deep learning models. For the best-performing MLP on Joint Positions, test accuracy dropped from **73.23%** (Without LDA) to **48.84%** (With LDA). This suggests that the linear reduction of LDA might collapse non-linear relationships that the MLP effectively captures. On the other hand, LDA improved the "Filter" traditional model in most cases, such as in the Combined set where test accuracy rose from **48.32%** to **58.19%**.

9. Feature Sensitivity

Joint Positions proved to be the most robust feature set, yielding the highest overall test accuracy (73.23%). **Euler Angles** typically resulted in lower performance (peak test accuracy of 69.08% for the Wrapper model). Interestingly, the **Combined** set did not significantly outperform Positions alone for the best models, suggesting that Joint Positions already contain the most discriminative information for this classification task.

The Combined feature set serves as the best example of LDA's utility. While it performed poorly in its raw state for traditional classifiers due to feature redundancy, the LDA-transformed version corrected these inefficiencies, yielding a more robust and generalized performance across subjects.

By combining spatial (Positions) and rotational (Euler) data, the model gains a more holistic view of the gesture. LDA ensures that the most relevant components of both feature types are prioritized, making the classification less sensitive to the specific way a subject might favour one type of movement over another.

While the MLP still struggled with the linear constraints of LDA on this set, the fact that three out of four models (Filter, Wrapper, and CNN) clustered around the **60% accuracy** mark with

LDA indicates that the Combined set provides a consistently reliable baseline when paired with discriminative projection

10. Overfitting

Traditional models exhibited severe overfitting across all feature sets. For instance, the Traditional w/ Wrapper model on Positions reached **97.42%** training accuracy but only **72.43%** on the test set. The CNN models showed more generalized behaviour, with smaller gaps between training and testing splits (e.g., **74.39%** train vs. **66.55%** test for Positions without LDA).

11. Generalization Across Subjects (Data Discussion)

The standard deviation across LOSO folds (avg. 4-6%) suggests that while the models are generally stable, subject-specific "outliers" exist. These individuals perform gestures with enough variation to significantly deviate from the group mean, representing the primary challenge for real-world deployment.

The standard deviation for testing accuracy ranges roughly between **3.39%** and **8.65%**. The high standard deviation in the MLP Positions (Without LDA) test set (5.06%) indicates that certain subjects in the LOSO cross-validation performed significantly differently than the mean. Subject-specific variability suggests that some participants may have performed motions uniquely, making them "outliers" relative to the generalized model training.

12. Conclusion

The **Joint Positions** are the most discriminative individual feature set, and **MLP** is the most capable architecture for handling the 46-class unmerged task. For systems where computational efficiency is prioritized, a **Traditional Wrapper with LDA on Combined features** offers the best balance of robustness and simplicity.

The experimental results indicate that while traditional models achieve near-perfect training accuracy, they fail to generalize as well as Deep Learning models, specifically the MLP, which achieved the highest test performance on Joint Position data. LDA is a useful dimensionality

reduction tool for simple traditional classifiers but inhibits the performance of more complex non-linear models like the MLP. For future work, focusing on the robustness of Joint Position features with improved regularization for Deep Learning models is recommended to bridge the generalization gap.

Links of Importance

Github Repo : https://github.com/deepk2001/Human_Motion_Classifier

Data Excel for metrics : [PRML_Classification_project_2_metrics.xlsx](#)

Email : djk6507@psu.edu